Maximal topological complexity of connected sums of aspherical manifolds

Kaiwen Jin
yui.ui.siki@gmail.com

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Abstract

Let M and N be closed, connected, oriented aspherical manifolds of the same dimension $n \geq 3$. We prove that their connected sum has ordinary unreduced topological complexity $2n+1$. Thus joining two aspherical summands always attains the dimensional upper bound, although the connected sum is not required to be aspherical. The proof detects the two ordered cross products of the summands' fundamental classes in a quotient of the left-right join of their free-product fundamental group. A decomposition into factor and cyclic cosets shows that these mixed classes survive, and their survival obstructs a motion planner with $2n$ domains.

1 Introduction

The topological complexity of a path-connected space measures how many continuous rules are needed to choose a path between arbitrary endpoints. Following Farber [1], we use the unreduced convention: $\mathrm{TC}(X)$ is the least number of open sets covering $X \times X$ on which

$$e : X^{[0,1]} \longrightarrow X \times X, \qquad e(\gamma) = (\gamma(0), \gamma(1)),$$

has continuous sections. The path space has the compact-open topology. For a connected n-manifold the dimensional upper bound is $2n+1$.

Connected sums offer a natural test of how motion-planning complexity changes under gluing. For aspherical summands the fundamental group of the sum is a free product, and the corresponding classifying space is a wedge. The free-product formula of Dranishnikov and Sadykov [2] gives $\mathrm{TC}(M \vee N) = 2n+1$ when M, N are closed oriented aspherical n-manifolds, $n \geq 3$. This computation does not give a lower bound for the connected sum: the fundamental-group isomorphism induced by pinching the neck does not make the connected sum aspherical.

Neofytidis [3, Question 5.5(b)] asks whether $\mathrm{TC}(M\#N) = 2n+1$ always holds in this setting. His Corollary 1.3 establishes the maximum for certain four-dimensional sums with a negatively curved summand. The revised work of Daundkar, Santhanam and Thandar [4, Proposition 5.3] computes the value 7 for wedges of closed oriented aspherical three-manifolds and gives bounds for specified connected sums. The question for connected sums therefore requires an obstruction that survives on their own fundamental class.

Theorem 1.1. *For every pair of closed, connected, oriented aspherical manifolds M, N of the same dimension $n \geq 3$,*

$$\mathrm{TC}(M\#N) = 2n+1.$$

The theorem includes summands with nontrivial center and aspherical rational homology spheres. There is no curvature or intermediate cohomology hypothesis. In particular, it gives $\mathrm{TC}(M\#M) = 7$ for every closed oriented aspherical three-manifold M, including the torus and the examples in [3, Question 5.10].

The argument uses the diagonal covering and left-right join construction of Farber, Grant, Lupton and Oprea [5]. We study its orbit space rather than transfer the wedge computation directly. In the free product, every simplex with nontrivial stabilizer has all its group labels in a single factor coset or a hyperbolic cyclic coset. Distinct such cosets intersect in at most one label. Their Borel constructions can contribute only the two same-summand products in top homology. The two ordered mixed products consequently survive in the orbit space, where a planner with $2n$ domains would force them to vanish.

We first describe this homological obstruction, then prove the coset decomposition and its homology consequence. The final section applies it to the connected-sum fundamental class. All homology below has coefficients $\mathbb{Q}$. Products and infinite CW constructions use the compactly generated convention; see [7, Chapters 5 and 10].

2 A homological obstruction from the diagonal covering

Let G be a discrete group and put $P = G \times G$. For a nonempty set S, let

$$E(S) = S * S * S * \cdots$$

be the join of countably many copies of S. Its vertices are (i, s), $i \geq 1$, and a simplex contains at most one vertex at each level i. We give it the weak CW topology. This infinite join is contractible: a sphere map has compact image in a finite subcomplex, and that image can be coned to a vertex at an unused level. The CW Whitehead theorem then applies.

Set $E = E(G)$. The left-right action

$$(u, v) \cdot (i, g) = (i, ugv^{-1})$$

makes E a P-CW complex. An element that stabilizes a simplex fixes it pointwise because it preserves the levels. Write E_m for the join of the first m copies of G, of dimension $m - 1$. These are the join models in [5, Sections 2.5–2.9].

Choose a contractible free P-CW complex EP. For a P-space W write $W_{hP} = (EP \times W)/P$, with diagonal action, for its Borel construction. The maps

$$\pi : E_{hP} \longrightarrow BP = EP/P, \qquad \rho : E_{hP} \longrightarrow E/P$$

forget the E and EP coordinates, respectively. The first is an associated bundle with contractible fiber, so it induces a homology isomorphism. We shall use

$$\tau_d = \rho_*(\pi_*)^{-1} : H_d(P) \longrightarrow H_d(E/P), \tag{1}$$

identifying group homology with the homology of BP.

Lemma 2.1. *Let Y be a closed connected manifold with fundamental group G, and let $c_Y : Y \to BG$ be a classifying map. If $\mathrm{TC}(Y) \leq m$, then for every $d \geq m$ and $z \in H_d(Y \times Y)$,*

$$\tau_d\big((c_Y \times c_Y)_* z\big) = 0.$$

Here $H_d(BG \times BG)$ is identified with $H_d(P)$ through the classifying homotopy equivalence inducing the identity on P.

Proof. Let $\widetilde{Y} \to Y$ be the universal cover. Lifting a path and taking its two lifted endpoints modulo simultaneous deck transformations gives a continuous map from $Y^{[0,1]}$ to the diagonal covering

$$q : (\widetilde{Y} \times \widetilde{Y})/\Delta G \longrightarrow Y \times Y,$$

over the endpoint map. The map is independent of the chosen lift. Continuity follows from local trivializations and path lifting; the compact interval can be covered by finitely many trivializing neighborhoods along each path. Thus every local motion planner gives a local section of q.

The covering q is associated to the principal P-cover $\widetilde{Y} \times \widetilde{Y} \to Y \times Y$ with fiber G and action ugv^{-1}; compare [5, proof of Theorem 2.1]. Choose a partition of unity for the m section domains, with each closed support contained in its domain. The weighted join of the local sections is a section of the associated bundle with fiber E_m. It is continuous in the weak join topology: near a point, each nonzero section label is locally constant in a covering trivialization, while the weights whose supports miss that point vanish on a neighborhood. The section is equivalently a P-equivariant map

$$f : \widetilde{Y} \times \widetilde{Y} \longrightarrow E_m.$$

Choose a P-equivariant classifying map $j : \widetilde{Y} \times \widetilde{Y} \to EP$. The map

$$b : Y \times Y \longrightarrow E_{hP}, \qquad b([y_0, y_1]) = [j(y_0, y_1), f(y_0, y_1)]$$

projects to the classifying map of the principal P-cover, homotopic to $c_Y \times c_Y$ after the stated identification. Hence $\pi_* b_* = (c_Y \times c_Y)_*$. Its composition with ρ factors through E_m/P, a CW complex of dimension $m-1$. It vanishes on H_d for $d \geq m$, which proves the lemma by (1). The universal cover of Y was not assumed contractible. □

3 Cosets in the free-product join

For the remainder of the argument let $G = A * B$, where A, B are nontrivial torsion-free groups. The factors are distinguished by their inclusions even if they are abstractly isomorphic. The free-product tree has vertices $G/A \sqcup G/B$ and an edge g joining gA to gB, with G acting by left translation. An element is called *hyperbolic* if it fixes no vertex of this tree.

Lemma 3.1. *The conjugates of A, B, together with the maximal cyclic subgroups generated by hyperbolic elements of the free-product tree, satisfy:*

1. *two distinct subgroups in this collection intersect trivially;*
2. *the normalizer of every subgroup in the collection is itself;*
3. *the centralizer of a nonidentity element is contained in one subgroup in the collection.*

Proof. Reduced alternating words show that the free-product graph is connected and has no circuits. Edge stabilizers are trivial; vertex stabilizers are conjugates of the factors; the action preserves the two vertex types and has no inversions.

A nonidentity element that fixes a vertex fixes no other vertex, since otherwise it fixes an edge in the segment between them. Its centralizer preserves its unique fixed vertex and lies in that vertex stabilizer. Distinct vertex stabilizers therefore intersect trivially. A normalizer of a vertex stabilizer preserves that vertex: choose any nonidentity element of the stabilizer and use uniqueness of its fixed vertex. This proves the normalizer assertion for factors.

A hyperbolic element g translates a unique axis, a bi-infinite geodesic. To see this, choose a vertex v minimizing $\ell = d(v, gv) > 0$. If $[v, gv]$ and $[gv, g^2v]$ backtrack at gv, let w be the neighbor of v toward gv. Then gw is the neighbor of gv toward v. For $\ell \geq 2$ this gives $d(w, gw) = \ell - 2$, contrary to minimality. For $\ell = 1$, the element exchanges the endpoints of an edge, contrary to the absence of inversions. The translates therefore concatenate without backtracking and give the axis, characterized by minimum displacement, which also establishes its uniqueness.

The stabilizer of an axis acts faithfully on it, since an element in the kernel fixes an edge. A reflection has square in this kernel and would be an element of order two. Torsion freeness excludes it. The axis stabilizer is thus a nonzero subgroup of the translations of a simplicial line, so is infinite cyclic. It is the centralizer of every nonidentity translation on that axis and the maximal cyclic subgroup containing it.

Two such cyclic groups sharing a nonidentity element have the same axis and equal its stabilizer. If hCh^{-1} meets C nontrivially, their axes agree, so h belongs to the axis stabilizer C. This proves malnormality and the normalizer assertion for these cyclic groups. They meet no vertex stabilizer nontrivially because their nonidentity elements are translations. All assertions follow. □

Let $\mathcal{H}$ contain A, B and one representative of each conjugacy class of maximal hyperbolic cyclic subgroups. Call a subset

$$D = uHv^{-1} \subset G, \qquad H \in \mathcal{H}, \quad u, v \in G,$$

a *block*, and identify duplicate subsets. Let $\mathcal{B}$ be the set of blocks, and put

$$Z = \bigcup_{D \in \mathcal{B}} E(D) \subset E.$$

For $g \in G$, write $F_g = E(\{g\})$. This is a contractible infinite simplex, consisting of all simplices with the same group label g at their different levels. Every F_g lies in Z.

Lemma 3.2. *Two distinct blocks intersect in at most one group element. Every cell of E outside Z has trivial P-stabilizer. The setwise stabilizer of the block H is $H \times H$; it acts transitively on the labels of that block, with stabilizer ΔH at 1. The stabilizer of the label 1 in all of G is ΔG.*

Proof. A block is a left coset of a conjugate of H. If two blocks contain distinct g, h, then $g^{-1}h$ is a nonidentity element of both associated subgroups. Lemma 3.1 identifies the subgroups, and the common element g identifies their cosets.

For a simplex with nontrivial stabilizer, send one label to 1 by the left-right action. A stabilizer element then has the form (w, w), where w centralizes all its labels. A nonidentity such w has centralizer in a factor conjugate or an axis cyclic group, by Lemma 3.1. All the labels lie in that subgroup, so the original simplex lies in one block. Constant-label simplices also lie in Z.

If $uHv^{-1} = H$, then $uv^{-1} = h \in H$. Substituting $u = hv$ shows that v normalizes H, so $u, v \in H$. The converse and the asserted transitivity are immediate from the action. Stabilizing the label 1 means $u = v$, giving the two diagonal subgroups as stated. □

4 The mixed top classes survive

We first record the relative comparison used to pass from Borel homology to orbit homology.

Lemma 4.1. *For the pair (E, Z) just constructed, the projection induces an isomorphism*

$$H_*(E_{hP}, Z_{hP}) \longrightarrow H_*(E/P, Z/P).$$

Proof. Filter the pairs by Z together with the cell skeleta of E. By Lemma 3.2, each relative cell orbit is free. For a k-cell its Borel relative piece is $EP \times (D^k, S^{k-1})$, and its orbit piece is (D^k, S^{k-1}). Their projection is an isomorphism on relative homology because EP is contractible.

For completeness, apply the same filtration to the mapping cone of the cellular chain map induced by projection. Each associated graded complex is a direct sum of mapping cones of the free-cell projections above, hence is acyclic. A cycle of fixed total degree has finite cellular support. Its component in the largest filtration degree is a boundary in that graded complex. Subtracting a lift of that boundary lowers the largest degree. After finitely many repetitions the cycle is filled. The filtration is bounded below, and the cell degrees and EP-chain degrees are nonnegative, so no unbounded filtration enters a fixed total degree. The mapping cone is acyclic, proving the homology isomorphism. □

Assume now that A, B admit classifying spaces with homology zero in degrees greater than n, where $n \geq 3$. In $H_{2n}(P)$ define

$$V = \operatorname{im} H_{2n}(A \times A) + \operatorname{im} H_{2n}(B \times B),$$

where the maps are induced by the two inclusions into $G \times G$.

Proposition 4.2. *The image of $H_{2n}(Z_{hP})$ in $H_{2n}(P)$ is contained in V.*

Proof. The block intersections give a P-equivariant pushout

$$\begin{array}{ccc} \bigsqcup_{D\in\mathcal{B}} \bigsqcup_{g\in D} F_g & \longrightarrow & \bigsqcup_{D\in\mathcal{B}} E(D) \\ \downarrow & & \downarrow \\ \bigsqcup_{g\in G} F_g & \longrightarrow & Z. \end{array} \tag{2}$$

Indeed, a simplex of Z with two distinct labels lies in exactly one block. Distinct blocks are identified precisely along the F_g they share. The top arrow is a CW cofibration, since in each block the union of its constant-label subcomplexes is a subcomplex. Thus this is a homotopy pushout.

One can take its explicit equivariant mapping-cylinder model before forming Borel constructions. Products with EP and passage to P-orbits commute with these identifications, and the subcomplex inclusion remains a cofibration. The result is an ordinary homotopy pushout. The blocks have one P-orbit for each $H \in \mathcal{H}$, with stabilizer $H \times H$. Incidences within a representative block form one orbit with stabilizer ΔH, and all labels form one orbit with stabilizer ΔG. Since the joins in the three input spaces are contractible, their Borel homotopy types are

$$\begin{array}{ccc} \bigsqcup_{H\in\mathcal{H}} B(\Delta H) & \longrightarrow & \bigsqcup_{H\in\mathcal{H}} B(H \times H) \\ \downarrow & & \downarrow \\ B(\Delta G) & \longrightarrow & Z_{hP}. \end{array} \tag{3}$$

Only these homotopy types are asserted in (3). Their maps to BP are the classifying maps of the corresponding subgroup inclusions, up to conjugation.

Put $d = 2n$. For $H = A, B$, the assumed homology bound gives $H_{d-1}(B\Delta H) = 0$, because $2n - 1 > n$. For the remaining H, the classifying space BH is a circle, so the same vanishing holds. Also $H_d(B\Delta G) = 0$: a classifying space for the free product is $BA \vee BB$, with no homology above n. Mayer–Vietoris for (3) therefore gives a surjection

$$\bigoplus_{H \in \mathcal{H}} H_d(H \times H) \longrightarrow H_d(Z_{hP}).$$

The cyclic terms vanish in degree $d \geq 6$, since their product classifying spaces are two-tori. The only possible contributions come from $A \times A$ and $B \times B$, with exactly the inclusion maps defining V. Inner conjugation induces a homotopically trivial change of a classifying-space map and does not enlarge their images. This proves the proposition. □

Proposition 4.3. *If $z \in H_{2n}(P)$ does not belong to V, then $\tau_{2n}(z) \neq 0$.*

Proof. The exact homology sequence of (E_{hP}, Z_{hP}), together with Proposition 4.2, shows that $(\pi_*)^{-1}z$ has a nonzero relative image. The commutative diagram

$$\begin{array}{ccc} H_{2n}(E_{hP}) & \longrightarrow & H_{2n}(E_{hP}, Z_{hP}) \\ \downarrow{\scriptstyle \rho_*} & & \downarrow{\scriptstyle \cong} \\ H_{2n}(E/P) & \longrightarrow & H_{2n}(E/P, Z/P) \end{array}$$

has an isomorphism on the right by Lemma 4.1. Thus $\rho_*(\pi_*)^{-1}z$ has a nonzero relative image as well, and is nonzero in absolute homology. □

5 The connected-sum fundamental class

Proof of Theorem 1.1. Put $X = M\#N$, $A = \pi_1(M)$, $B = \pi_1(N)$. Closed topological manifolds have finite CW type, and the fundamental group of an aspherical finite-dimensional CW complex is torsion free; see [6, Theorem 5.3 and Lemma 3.1]. Each factor is nontrivial: a closed oriented contractible manifold would have both zero top homology and a nonzero fundamental class. The free product $G = A * B$ is torsion free, since an element of finite order is conjugate into a factor. In reduced-word terms, every other element is conjugate to a cyclically reduced alternating word of length at least two, whose positive powers have increasing length.

The wedge $M \vee N$ is a classifying space for G. On based CW models its universal cover is a tree of the contractible universal covers of the summands. Attaching points are cofibrations. A finite subtree of contractible vertex spaces is contractible by successive homotopy pushouts. A sphere map into the entire cover has compact image in a finite CW subcomplex, so meets only finitely many pieces and is null-homotopic in a finite subtree. Whitehead's theorem makes the full cover contractible.

For $n \geq 3$, pinching the connecting sphere gives a classifying map

$$c : X \longrightarrow M \vee N \simeq BG$$

inducing the free-product fundamental-group isomorphism. This is the usual connected-sum pinch map; compare the proof of Lemma 3.2 in [6]. One may first collapse the neck and then identify each punctured summand with its boundary collapsed with the closed summand up to homotopy. The latter operation collapses a contractible closed-ball cofibration.

Let $a, b \in H_n(BG)$ be the separate images of $[M]$ and $[N]$. Both projections of c onto a summand have degree $+1$: away from the removed ball each is the oriented identity near a point with a single preimage, hence sends its local orientation class to the positive generator. The two projections determine the top homology of the wedge, so

$$c_*[X] = a + b.$$

The rational Künneth theorem [7, Chapter 17, Section 2] gives

$$H_{2n}(P) = \mathbb{Q}\{a \times a,\ a \times b,\ b \times a,\ b \times b\}, \qquad V = \mathbb{Q}\{a \times a,\ b \times b\},$$

since the homology of each summand vanishes above n. Naturality of the cross product yields

$$(c \times c)_*[X \times X] = a \times a + a \times b + b \times a + b \times b \notin V.$$

The two mixed terms are independent ordered coordinates. They remain distinct also when $M = N$ or n is odd.

Proposition 4.3 now gives

$$\tau_{2n}\big((c \times c)_*[X \times X]\big) \neq 0.$$

If $\mathrm{TC}(X) \leq 2n$, Lemma 2.1 gives the opposite conclusion in degree $2n$. Hence $\mathrm{TC}(X) \geq 2n + 1$. Farber's dimension bound [1, Theorem 4] gives the reverse inequality, completing the proof. □

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